

# Op Amp: DAC, ADC, & Active Filtering Data Collection

Table 1: First-Order Active Filtering

|                      | Low Pass     |              | High Pass    |              | Band Pass    |              |
|----------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| First Order          | <i>Calc.</i> | <i>Meas.</i> | <i>Calc.</i> | <i>Meas.</i> | <i>Calc.</i> | <i>Meas.</i> |
| $F_{C_{LOW}}$        |              |              | $20_{HZ}$    |              | $20_{HZ}$    |              |
| $F_{C_{HIGH}}$       | $20K_{HZ}$   |              |              |              | $20K_{HZ}$   |              |
| $F_{Octave}$         |              |              |              |              |              |              |
| $F_{2\_Octave}$      |              |              |              |              |              |              |
| $F_{Decade}$         |              |              |              |              |              |              |
| $F_{2\_Decade}$      |              |              |              |              |              |              |
| $V_{pp_{PassBand}}$  |              |              |              |              |              |              |
| $V_{pp_{FC}}$        |              |              |              |              |              |              |
| $V_{pp_{Octave}}$    |              |              |              |              |              |              |
| $V_{pp_{2\_Octave}}$ |              |              |              |              |              |              |
| $V_{pp_{Decade}}$    |              |              |              |              |              |              |
| $V_{pp_{2\_Decade}}$ |              |              |              |              |              |              |
| $dB_{PassBand}$      |              |              |              |              |              |              |
| $dB_{FC}$            |              |              |              |              |              |              |
| $dB_{Octave}$        |              |              |              |              |              |              |
| $dB_{2\_Octave}$     |              |              |              |              |              |              |
| $dB_{Decade}$        |              |              |              |              |              |              |
| $dB_{2\_Decade}$     |              |              |              |              |              |              |

Table 2: Second-Order Active Filtering

|                      | Low Pass     |              | High Pass    |              | Cascaded     |              |
|----------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Second Order         | <i>Calc.</i> | <i>Meas.</i> | <i>Calc.</i> | <i>Meas.</i> | <i>Calc.</i> | <i>Meas.</i> |
| $F_{C_{LOW}}$        |              |              | $20_{HZ}$    |              | $20_{HZ}$    |              |
| $F_{C_{HIGH}}$       | $20K_{HZ}$   |              |              |              | $20K_{HZ}$   |              |
| $F_{Octave}$         |              |              |              |              |              |              |
| $F_{2\_Octave}$      |              |              |              |              |              |              |
| $F_{Decade}$         |              |              |              |              |              |              |
| $F_{2\_Decade}$      |              |              |              |              |              |              |
| $V_{pp_{PassBand}}$  |              |              |              |              |              |              |
| $V_{pp_{FC}}$        |              |              |              |              |              |              |
| $V_{pp_{Octave}}$    |              |              |              |              |              |              |
| $V_{pp_{2\_Octave}}$ |              |              |              |              |              |              |
| $V_{pp_{Decade}}$    |              |              |              |              |              |              |
| $V_{pp_{2\_Decade}}$ |              |              |              |              |              |              |
| $dB_{PassBand}$      |              |              |              |              |              |              |
| $dB_{FC}$            |              |              |              |              |              |              |
| $dB_{Octave}$        |              |              |              |              |              |              |
| $dB_{2\_Octave}$     |              |              |              |              |              |              |
| $dB_{Decade}$        |              |              |              |              |              |              |
| $dB_{2\_Decade}$     |              |              |              |              |              |              |